

TWPT

Probability – 01

- If for two events A and B , $P(A + B) = P(A) + P(B)$, then these events are
(a) mutually exclusive (b) independent
(c) dependent (d) None of these
- In two throws of a coin (or a throw of two coins), the probability of getting atleast one head is
(a) $1/2$ (b) $3/4$ (c) $1/3$ (d) $1/4$
- In a throw of three coins (or 3 throws of a coin), the probability of getting 3 heads is
(a) $1/8$ (b) $3/8$ (c) $5/8$ (d) $1/4$
- Two dice are thrown thrice. The probability that first throw shows 10, second 11 and third 12 will be
(a) $1/216$ (b) $1/432$ (c) $1/7776$ (d) $1/648$
- In 4 throws of a die, the probability of getting 5 to 6 atleast once is
(a) $65/81$ (b) $16/81$ (c) $2/3$ (d) $1/3$
- A card is drawn from a well shuffled pack of 52 cards. Its probability of being an ace or a king or a queen or a jack is
(a) $1/13$ (b) $2/13$ (c) $3/13$ (d) $4/13$
- One number is selected from first 120 natural numbers. The probability that it is divisible by 5 or 15 is
(a) $1/5$ (b) $1/8$ (c) $1/16$ (d) None
- Three urns contain 2 white and 3 black; 3 white and 2 black; and one white and one black balls. A ball is drawn from one of these three urns. The probability that it is a white ball will be
(a) $1/5$ (b) $2/5$ (c) $1/2$ (d) $4/5$
- A man can kill a bird once in three shots. On this assumption he fires three shots. The probability that the bird is killed is
(a) $8/27$ (b) $19/27$ (c) $7/27$ (d) $11/27$
- If from a lottery of 30 tickets marked 1, 2, 3 30 four tickets be drawn. The chance that those marked 1 and 2 are among them is
(a) $1/145$ (b) $2/145$ (c) $4/145$ (d) None
- A bag contains 5 red and 3 white balls. A man draws two balls one by one. The probability that one drawing ball is white and other is red will be
(a) $15/56$ (b) $30/56$
(c) $30/64$ (d) $26/64$
- 3 letters are placed in 3 envelopes randomly. The probability that no letter goes to the right envelope is
(a) $1/6$ (b) $1/2$ (c) $1/3$ (d) None
- 3 letters are placed in 3 envelopes randomly. The probability that only one letter goes in right envelope is
(a) $1/2$ (b) $1/3$ (c) $1/6$ (d) $2/3$
- A bag contains 3 white and 3 black balls. Balls are drawn one by one without replacing them in the bag. The probability that drawing ball will be in alternate colours is
(a) $1/10$ (b) $5/21$ (c) $1/2$ (d) None
- For solving a problem, odds against to A are 4 : 3 and odds in favour to B are 7 : 5. The probability that the problem will not be solved is
(a) $16/21$ (b) $5/21$ (c) $43/84$ (d) $45/84$
- Three children are chosen at random from a group of 5 boys and 3 girls. The probability that atleast one girl will be in the selection is
(a) $5/28$ (b) $15/28$ (c) $23/28$ (d) $8/28$
- An instrument works only when its components A , B , C are in order. In a year the probability of disordering of A , B , C are 0.15, 0.05 and 0.10 respectively. What is the probability that the instrument will be in disorder before the end of the year
(a) 0.72675 (b) 0.27325
(c) 0.85 (d) 0.95
- The probability that a boy will pass an examination is $3/5$ and that of a girl is $2/5$. The probability that atleast one will pass the examination is
(a) $4/25$ (b) $6/25$ (c) $9/25$ (d) $19/25$
- The probabilities that three boys will pass an examination are $1/6$, $1/4$ and $1/3$ respectively. The probability that exactly one boy will pass the examination is :
(a) $31/72$ (b) $7/12$ (c) $41/72$ (d) $11/12$
- If $\frac{1-3p}{2}$, $\frac{1+4p}{3}$, $\frac{1+p}{6}$ are the probabilities of three mutually exclusive and exhaustive events, then the set of all values of p is
(a) $(0,1)$ (b) $[-1/4, 1/3]$ (c) $(0, 1/3)$ (d) $(0, \infty)$
- Three cards are drawn from a pack of 52 playing cards. The probability that they are of the same colour is
(a) $1/17$ (b) $2/17$ (c) $3/17$ (d) $4/17$
- Three dice are thrown together. The probability that all will show even number is
(a) $3/216$ (b) $9/216$
(c) $27/216$ (d) None of these
- Three letters are chosen from the letters of the word 'SLEEPER'. The probability of including atleast two E's is
(a) $13/35$ (b) $12/35$ (c) $1/35$ (d) $14/35$
- Three bulbs are chosen from 15 bulbs. If 5 of them are fused then the probability that no bulb is fused will be
(a) $49/91$ (b) $51/91$ (c) $24/91$ (d) $59/91$
- The probabilities of solving a problem by A and B are $2/3$ and $3/4$ respectively. The probability of solving the problem by atmost one is
(a) $1/2$ (b) $1/4$ (c) $5/12$ (d) $2/3$
- From the records of a hospital, it is found that 20% patients died with the disease cancer. If two patients with cancer are admitted to hospital; then probability that atleast one patient will be cured, is

- (a) 16/25 (b) 24/25 (c) 9/25 (d) None
27. A bag contains two pairs of shoes (i.e. 4 shoes in all). Two shoes are drawn at random. The probability that it is a pair of proper shoes is
(a) 1/4 (b) 2/3 (c) 1/3 (d) None
28. A bag contains 5 white and 3 red balls. Balls are drawn one by one with replacement. The probability that there will be red ball in the fifth draw first time is
(a) 3/56 (b) 1/56 (c) 5/56 (d) None
29. There are 10 bulbs in a room, 4 of them are fused. The probability that the room will be lighted on pressing the switches of any three, is
(a) 1/30 (b) 29/30 (c) 3/5 (d) None
30. A bag contains 3 red and 2 white balls and B bag contains 2 red and 4 white balls. One ball is drawn from bag A and it is put in bag B. Then on ball is drawn from bag B. The probability that it will be a red ball is
(a) 13/35 (b) 9/35 (c) 4/35 (d) None
31. A and B play throwing 3 coins each. A wins if the number of heads is same in the throws of both A and B. The probability of winning A is
(a) 5/16 (b) 19/64 (c) 7/8 (d) None
32. A and B are two tennis players of equal standard. If they play 4 games, then probability that A wins exactly three games is
(a) 1/2 (b) 3/4 (c) 1/4 (d) None
33. A lottery has 50 tickets out of which 8 tickets have prizes. 5 tickets are selected at random. The probability that two prize tickets are included is
(a) 164/1081 (b) 165/1081
(c) 163/1081 (d) None of these
34. The odds in favour of escape of any enemy ship are 4 : 1 . The probability that atleast one ship out of three ships gets destroyed is
(a) 1/125 (b) 16/125 (c) 61/125 (d) 64/125
35. The probability that a three digit number formed with 1, 2, 3, 4 is divisible by 3 is
(a) 1/3 (b) 1/2 (c) 2/3 (d) 3/4
36. The probability that a four digit number formed with 1, 2, 3, 4, 5 is divisible by 3 is
(a) 1/3 (b) 1/4 (c) 1/5 (d) 2/3
37. The probability that a leap year will have 52 Sunday is
(a) 1 (b) 5/7 (c) 2/7 (d) None
38. Seven persons sit on a round table. The probability that two particular persons sit together is
(a) 1/3 (b) 2/3 (c) 1/6 (d) None
39. From a box containing 6 pairs of socks, 4 socks are drawn at random. The probability that they will contain atleast one complete pair is
(a) 17/33 (b) 1/33 (c) 16/34 (d) None
40. A has 3 shares in a lottery having 3 prizes and 6 blanks. B has one share in another lottery having 1 prizes and 2 blanks. Their probability of winning are in the ratio
(a) 16 : 7 (b) 7 : 16
(c) 1 : 3 (d) None of these
41. The odds against solving a problem by A are 4 : 3 and in favour of B are 7 : 5. The probability that only one will solve the problem is
(a) 16/21 (b) 5/21 (c) 43/84 (d) 41/84
42. Which of the following is false
(a) $P(AB) \leq P(A)$ (b) $P(A) \leq P(A+B)$
(c) $P(A+B) \leq P(A)+P(B)$ (d) None of these
43. Two cards are drawn from a pack of 52 cards. The probability that both are red or both are kings will be
(a) $\frac{325}{1326}$ (b) $\frac{1}{221}$ (c) $\frac{55}{221}$ (d) None
44. A die is thrown twice. When the sum of their digits be 6 then the probability that atleast one of them will show 4, is
(a) 1/6 (b) 2/5 (c) 3/5 (d) 1/2
45. If a random variate X has Poisson distribution such that $P(X=1) = P(X=2)$, then $P(X=4)$ is :
(a) $\frac{1}{2e^2}$ (b) $\frac{2}{3e^2}$ (c) $\frac{3}{4e^2}$ (d) $\frac{7}{8e^2}$
46. One hundred identical coins, each with probability p of showing up heads, are tossed once. If $0 < p < 1$ and the probability of heads showing on 50 coins is equal to that of heads showing on 51 coins, then p is equal to:
(a) 1/2 (b) 51/101
(c) 50/101 (d) 49/101
47. Three persons A, B and C are to speak at a function along with 5 others. If the persons speak in random order, the probability that A speaks before B and B speaks before C is :
(a) 1/12 (b) 1/8 (c) 1/6 (d) 1/4
48. An event A is independent of itself if and only if $P(A)$ is equal to :
(a) 0 (b) 1 (c) 1/2 (d) 0 or 1
49. A word is formed using all letters of the word 'SERIES'. The probability that all vowels will occur together is:
(a) 1/2 (b) 1/3 (c) 1/5 (d) 2/5
50. A set A has 4 elements. One- one onto functions are defined on A. If one function is selected from them then the probability that it satisfies $f(x) \neq x \forall x \in A$, will be:
(a) 11/24 (b) 9/24 (c) 15/24 (d) 17/24

ANSWER KEY : TWPT (Probability – 1)

1. (a)	2. (b)	3. (a)	4. (c)	5. (a)	6. (d)	7. (a)	8. (c)	9. (b)	10. (b)
11. (b)	12. (c)	13. (a)	14. (a)	15. (b)	16. (c)	17. (b)	18. (d)	19. (a)	20. (b)
21. (d)	22. (c)	23. (a)	24. (c)	25. (a)	26. (b)	27. (c)	28. (a)	29. (b)	30. (a)
31. (b)	32. (c)	33. (a)	34. (c)	35. (b)	36. (c)	37. (b)	38. (a)	39. (a)	40. (a)
41. (c)	42. (d)	43. (d)	44. (b)	45. (b)	46. (b)	47. (c)	48. (d)	49. (c)	50. (b)

impetus

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